

4.3.49

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Question

Write the equation of the lines for which $\tan \theta = \frac{1}{2}$, where θ is the inclination of the line, and

(a) y intercept $-\frac{3}{2}$

(b) x intercept 4

Solution

Variable	Value
A	$(0, -\frac{3}{2})$
m	$\frac{1}{2}$

Table: Variables Used

Solution

Let **B** be a point on the line

$$\mathbf{A} = \begin{pmatrix} 0 \\ -\frac{3}{2} \end{pmatrix} \quad (1)$$

$$\text{Let } \mathbf{M} = \begin{pmatrix} 1 \\ m \end{pmatrix} \quad (2)$$

$$\mathbf{M} = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} \quad (3)$$

$$(\mathbf{B} - \mathbf{A}) = \mathbf{M} \quad (4)$$

$$\mathbf{B} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (5)$$

Let eq of line be

$$\mathbf{n}^T \mathbf{x} = 1 \quad (6)$$

As **A** and **B** lie on the line

$$\mathbf{n}^T \mathbf{A} = 1 \quad (7)$$

$$\mathbf{n}^T \mathbf{B} = 1 \quad (8)$$

Solution

From eq (7) and (8)

$$\begin{pmatrix} 0 & -\frac{3}{2} \\ 1 & -1 \end{pmatrix} \mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (9)$$

In augmented matrix form

$$\left[\begin{array}{cc|c} 0 & -\frac{3}{2} & 1 \\ 1 & -1 & 1 \end{array} \right] \quad (10)$$

Swap R_1 and R_2

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & -\frac{3}{2} & 1 \end{array} \right] \quad (11)$$

Solution

$$R_2 \rightarrow -\frac{2R_2}{3}$$

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & -\frac{2}{3} \end{array} \right] \quad (12)$$

$$R_1 \rightarrow R_1 + R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{2}{3} \end{array} \right] \quad (13)$$

$$\mathbf{n} = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \end{pmatrix} \quad (14)$$

hence eq of line is

$$\mathbf{n}^T \mathbf{x} = 1 \quad (15)$$

$$\left(\frac{1}{3} \quad -\frac{2}{3} \right) \mathbf{x} = 1 \quad (16)$$

Graph

Refer to Figure

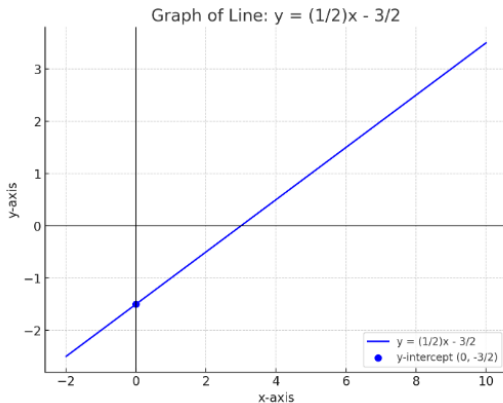


Figure:

Question

Write the equation of the lines for which $\tan \theta = \frac{1}{2}$, where θ is the inclination of the line, and

(a) y intercept $-\frac{3}{2}$

(b) x intercept 4

Solution

Variable	Value
A	$(4, 0)$
m	$\frac{1}{2}$

Table: Variables Used

Solution

Let **B** be a point on the line

$$\mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad (17)$$

$$\text{Let } \mathbf{M} = \begin{pmatrix} 1 \\ m \end{pmatrix} \quad (18)$$

$$\mathbf{M} = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} \quad (19)$$

$$(\mathbf{B} - \mathbf{A}) = \mathbf{M} \quad (20)$$

$$\mathbf{B} = \begin{pmatrix} 5 \\ \frac{1}{2} \end{pmatrix} \quad (21)$$

Let eq of line be

$$\mathbf{n}^T \mathbf{x} = 1 \quad (22)$$

As $\mathbf{A}$ and $\mathbf{B}$ lie on the line

$$\mathbf{n}^T \mathbf{A} = 1 \quad (23)$$

$$\mathbf{n}^T \mathbf{B} = 1 \quad (24)$$

Solution

From eq 23 and 24

$$\begin{pmatrix} 4 & 0 \\ 5 & \frac{1}{2} \end{pmatrix} \mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (25)$$

In augmented matrix form

$$\left[\begin{array}{cc|c} 4 & 0 & 1 \\ 5 & \frac{1}{2} & 1 \end{array} \right] \quad (26)$$

$$R_1 \rightarrow \frac{R_1}{4}$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{4} \\ 5 & \frac{1}{2} & 1 \end{array} \right] \quad (27)$$

$$R_2 \rightarrow R_2 - 5R_1$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{4} \\ 0 & \frac{1}{2} & -\frac{1}{4} \end{array} \right] \quad (28)$$

Solution

$$R_2 \rightarrow 2R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{4} \\ 0 & 1 & -\frac{1}{2} \end{array} \right] \quad (29)$$

$$\mathbf{n} = \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{2} \end{pmatrix} \quad (30)$$

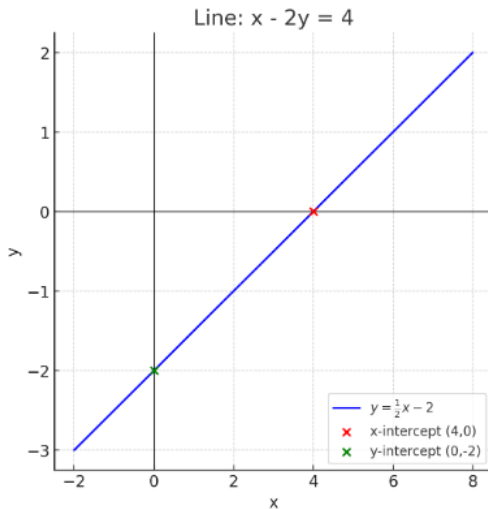
hence eq of line is

$$\mathbf{n}^T \mathbf{x} = 1 \quad (31)$$

$$\left(\frac{1}{4} \quad -\frac{1}{2} \right) \mathbf{x} = 1 \quad (32)$$

Graph

Refer to Figure



Python Code 1

```
1 import matplotlib.pyplot as plt
2 import numpy as np
3
4 # Define slope and intercept
5 m = 1/2
6 c = -3/2
7
8 # Create x values
9 x = np.linspace(-2, 10, 400)
10
11 # Equation of the line
12 y = m * x + c
13
14 # Plot the line
15 plt.figure(figsize=(8,6))
16 plt.plot(x, y, label=  $y = (1/2)x - 3/2$  , color= blue )
17
18 # Mark the y-intercept point
19 plt.scatter(0, c, color= blue , marker= 'o' , label=  $y$  )
```


Python Code 1

```
# Draw x and y axes
plt.axhline(0, color='black', linewidth=0.8)
plt.axvline(0, color='black', linewidth=0.8)

# Labels and title
plt.xlabel( x-axis )
plt.ylabel( y-axis )
plt.title( Graph of Line:  $y = (1/2)x - 3/2$  )
plt.legend()
plt.grid(True)

# Save the graph as PNG
plt.savefig( grapha , dpi=300)

# Show plot
plt.show()
```

Python Code 2

```
import matplotlib.pyplot as plt
import numpy as np

# Define the line:  $y = (1/2)x - 2$ 
x = np.linspace(-2, 8, 200)
y = 0.5 * x - 2

# Create plot
plt.figure(figsize=(6,6))
plt.plot(x, y, label=r  $y=\frac{1}{2}x-2$  , color=
        blue )

# Mark intercepts
plt.scatter([4], [0], color= red , label= x-intercept
        (4,0) , zorder=5)
plt.scatter([0], [-2], color= green , label= y-
        intercept (0,-2) , zorder=5)
```

```
# Axes
```

Python Code 2

```
# Labels
plt.xlabel( x )
plt.ylabel( y )
plt.title( Line:  $x - 2y = 4$  )
plt.legend()
plt.grid(True)

# Save and show
file_path = line_plot.png      # will save in current
    directory
plt.savefig(file_path)
plt.show()

print(f Plot saved as {file_path} )
```

C Code

```
#include <stdio.h>

// Print 2x3 matrix
void printMatrix(float mat[2][3]) {
    for(int i=0; i<2; i++) {
        for(int j=0; j<3; j++) {
            printf( "%6.2f", mat[i][j]);
        }
        printf( "\n ");
    }
    printf( "\n ");
}

// Gaussian Elimination
void gaussElimination(float mat[2][3]) {
    // Normalize first row
    float factor = mat[0][0];
    if (factor != 0) {
        for(int j=0; j<3; j++)
```

```
// Eliminate below
factor = mat[1][0];
for(int j=0; j<3; j++)
    mat[1][j] -= factor * mat[0][j];

// Normalize second row
factor = mat[1][1];
if (factor != 0) {
    for(int j=0; j<3; j++)
        mat[1][j] /= factor;
}

// Eliminate above
factor = mat[0][1];
for(int j=0; j<3; j++)
    mat[0][j] -= factor * mat[1][j];
}
```

Python and C Code

```
import ctypes
import numpy as np

# Load the shared object
lib = ctypes.CDLL( './line_solver.so' )

# Define function signatures
lib.gaussElimination.argtypes = [((ctypes.c_float * 3)
    * 2)]
lib.printMatrix.argtypes = [((ctypes.c_float * 3) * 2)
    ]

def to_c_matrix(py_mat):
    Convert Python 2x3 list into C float[2][3]
    c_mat = ((ctypes.c_float * 3) * 2)()
    for i in range(2):
        for j in range(3):
            c_mat[i][j] = py_mat[i][j]
    return c_mat
```

```
def to_py_matrix(c_mat):  
    Convert C float[2][3] back to Python list  
    return [[c_mat[i][j] for j in range(3)] for i in  
            range(2)]  
  
# Case (a): slope=1/2, y-intercept=-3/2  
mat1 = [[0, 1, -1.5],  
        [1, -0.5, 0]]  
  
c_mat1 = to_c_matrix(mat1)  
print( Case (a): before elimination: )  
lib.printMatrix(c_mat1)  
  
lib.gaussElimination(c_mat1)
```

Python and C Code

```
print( Case (a): after elimination: )
lib.printMatrix(c_mat1)
print( Python Matrix: , to_py_matrix(c_mat1))

# Case (b): slope=1/2, x-intercept=4
mat2 = [[4, 1, 0],
        [1, -0.5, 0]]

c_mat2 = to_c_matrix(mat2)
print( \nCase (b): before elimination: )
lib.printMatrix(c_mat2)

lib.gaussElimination(c_mat2)

print( Case (b): after elimination: )
lib.printMatrix(c_mat2)
print( Python Matrix: , to_py_matrix(c_mat2))
```